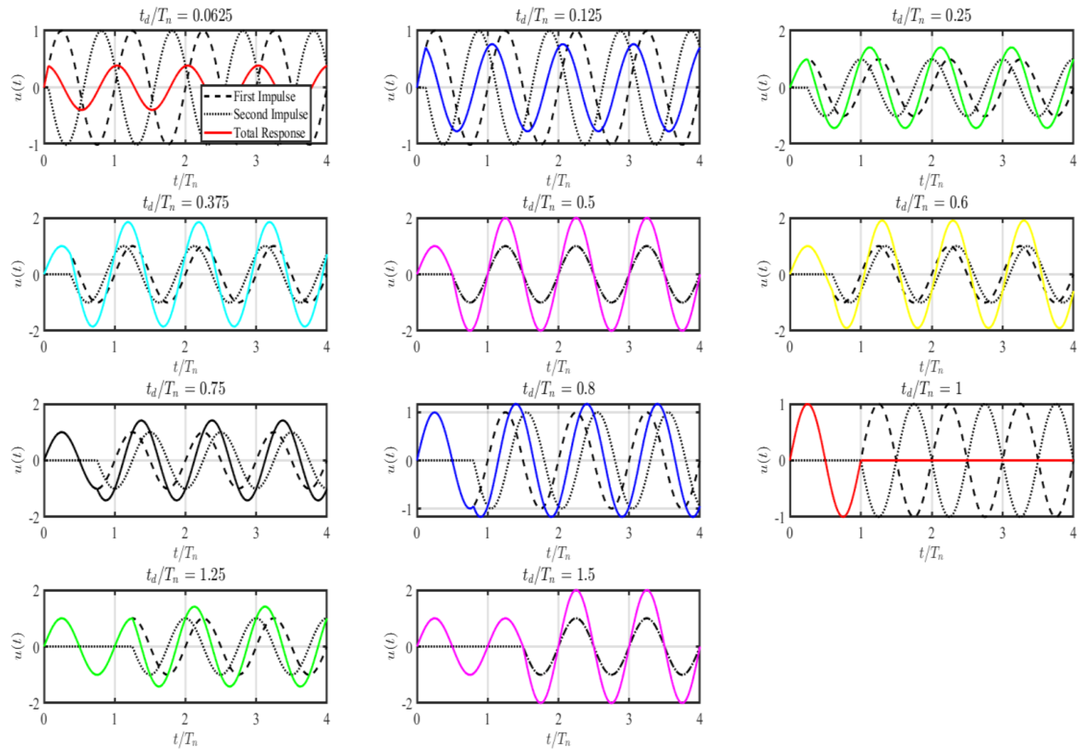
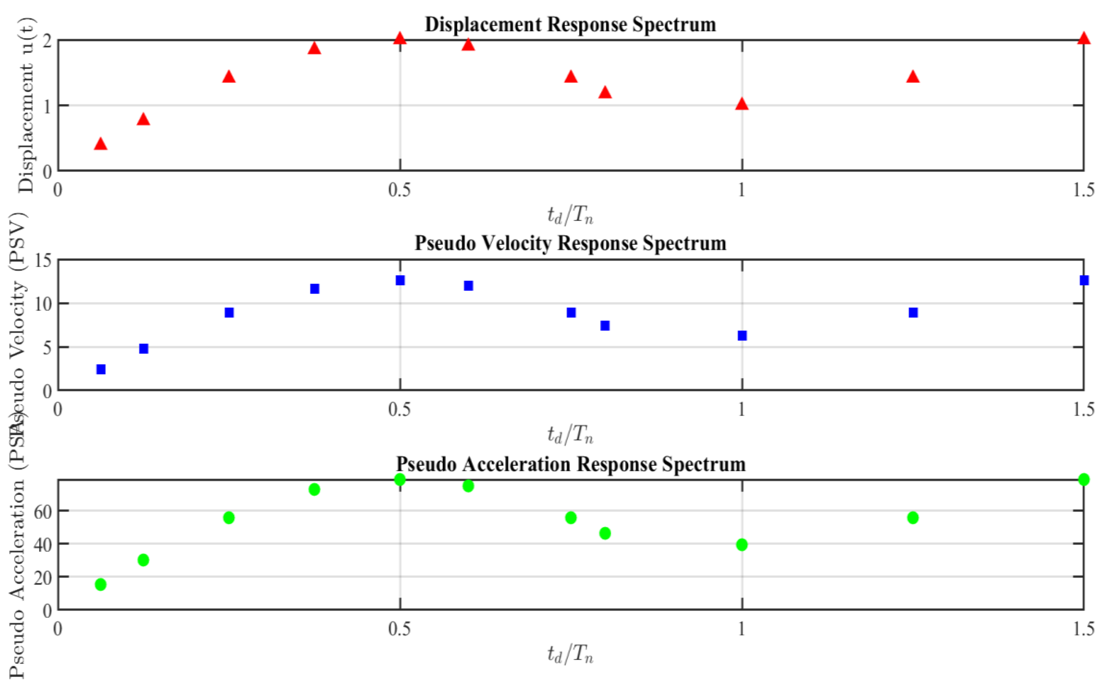
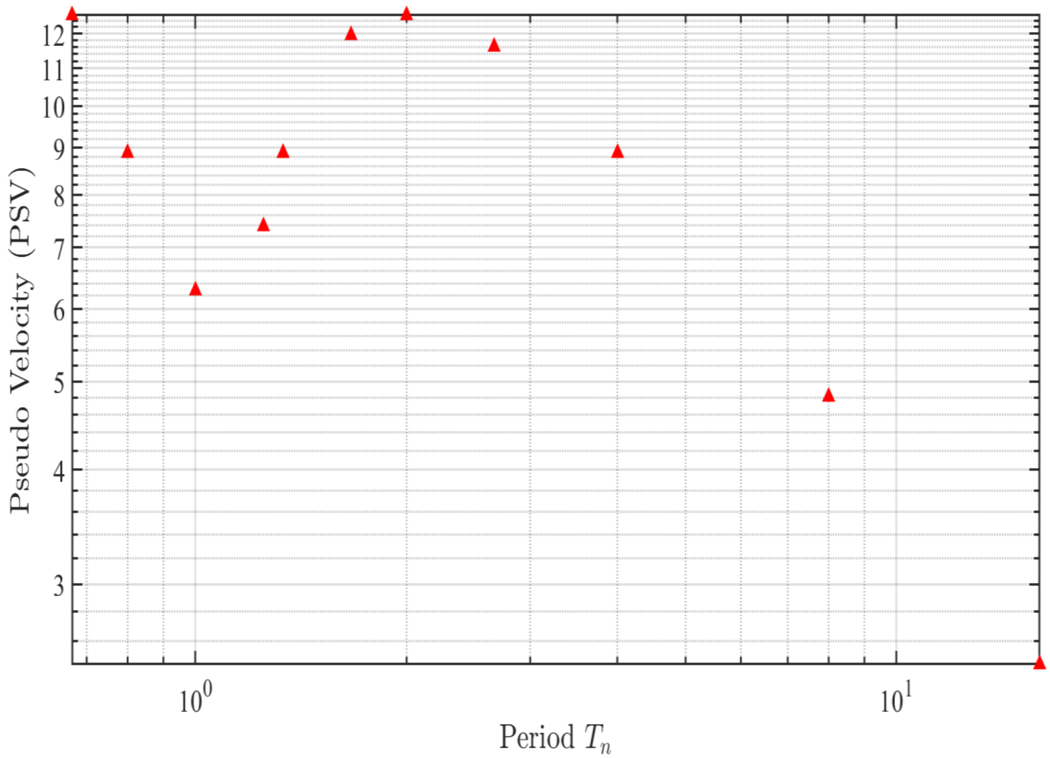
(1).


$t_d/T_n = 0.0625$
$t_d/T_n = 0.125$
$t_d/T_n = 0.25$
$t_d/T_n = 0.375$
$t_d/T_n = 0.5$
$t_d/T_n = 0.6$
$t_d/T_n = 0.75$
$t_d/T_n = 0.8$
$t_d/T_n = 1$
$t_d/T_n = 1.25$
$t_d/T_n = 1.5$
First Impulse
Second Impulse
Total Response
$u(t)$
t/T_n


Displacement Response Spectrum
Displacement u(t)
Pseudo Velocity Response Spectrum
Pseudo Velocity (PSV)
Pseudo Acceleration Response Spectrum
Pseudo Acceleration (PSA)
t_d/T_n


Pseudo Velocity (PSV)
12
11
10
9
8
7
6
5
4
3
10^0
10^1
Period T_n

(2).

2 (a) $m = \frac{W}{g} = \frac{100}{386} = 0.26$ lbs

$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{4}{0.26}} = 3.93$ rad/s.

$T_n = \frac{2\pi}{\omega_n} = 1.60$ sec

According to the scaled design spectrum to 0.5g PGA,
from $T_n = 1$ sec to $T_n = 4.72$ sec, $\frac{A}{g} = 0.9 (T_n)^{-1}$

So, $\frac{A}{g} = 0.562$

$D = \frac{A}{\omega_n^2} = \frac{0.562 * 386}{(3.93)^2} = 14.1$ in.

$V_{bo} = (A/g) * W = 0.562 * 100 = 56.2$ kips.

(b) $\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{8}{0.26}} = 5.55$ rad/s;

$T_n = \frac{2\pi}{\omega_n} = 1.13$ s.

$A/g = \cancel{0.8}.\ 0.795$

$D = \frac{A}{\omega_n^2} = \frac{0.8 * 386}{(5.55)^2} = 9.93$ in.

$V_{bo} = (\frac{A}{g}) * W = \cancel{0.8}\ 0.795 * 100 = 79.5$ kips,

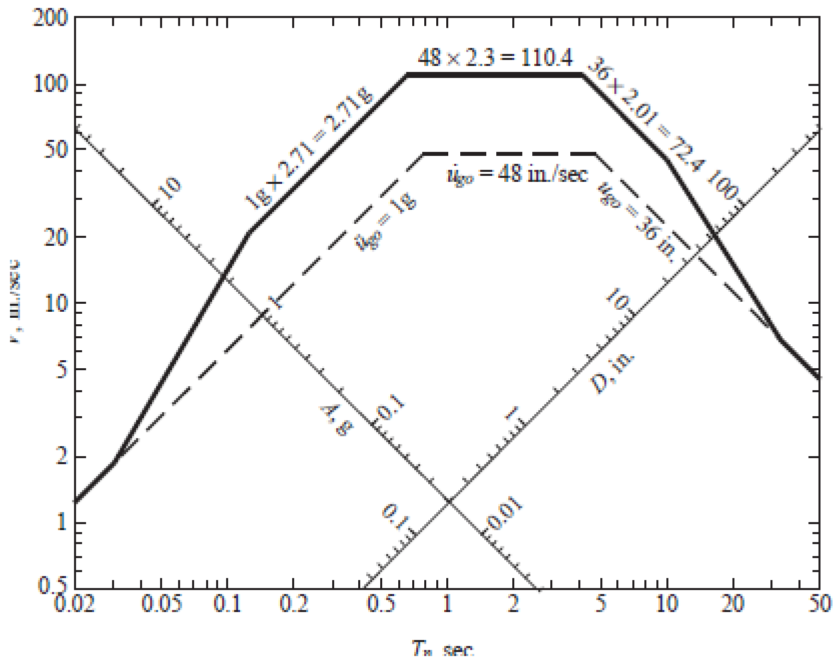
Stiffening the water tank shortens the nature (period
and decreases the design deformation, however this Increases
the base ~~stress~~ shear, i.e disadvantage.

For each system we compute $\omega_n = \sqrt{(k/m)}$ and $T_n = 2\pi/\omega_n$. For the computed T_n and $\zeta = 5\%$ we read one of D, V, or A from the design spectrum of below multiplied by 0.50; and compute the other two among D, V, and A. The peak deformation $u_o = D$ and the base shear is $V_{bo} = (A/g)w$. These results are summarized in the accompanying table.

System	ω_n sec^{-1}	T_n sec	V in/sec	D in.	A/g	V_{bo} kips
(a)	3.93	1.60	55.2	14.1	0.562	56.2
(b)	5.56	1.13	55.2	9.93	0.795	79.5

Comparing the response of systems (a) and (b), we observe that stiffening the tower shortens the natural period and reduces the design deformation but increases the base shear; the latter is a disadvantage.

The preceding comments are restricted to systems in the constant-V region of the spectrum.



(3).

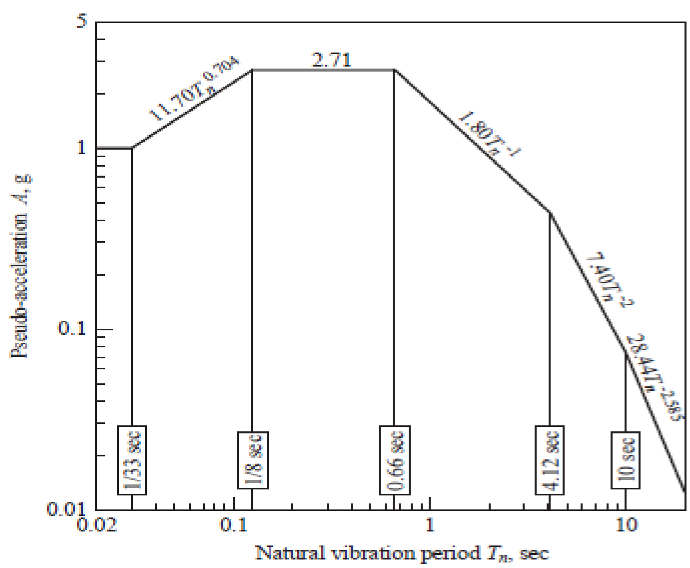
(a) Frame with rigid beam ($EI_1 = \infty$)

$k = (2 * 12 * E * I_c) / h^3 = (24 *(3* 10^3) * (10^4 / 12)) / (12 * 12)^3 = 20.09$ kips/in.

$\omega_n = \sqrt{(k / m)} = \sqrt{(20.09 /10/ 386)} = 27.85$ rad/sec

$T_n = 2\pi / \omega_n = 2\pi / 27.85 = 0.226$ sec

For $T_n = 0.226$ sec and $\zeta = 5\%$, from figure below scaled by 0.50 gives:



$A/g = 1.355$

$A/g = 1.355 \Rightarrow D = A/ (\omega_n)^2 = (1.355 * 386) / (27.85)^2 = 0.674$ in.

Design quantities:

Deformation, $u_o = 0.674$ in.

Lateral force, $f_{s0} = (A / g)w = 13.55$ kips

Bending moments at top and bottom of columns:

$M = (13.55 / 2) * h / 2 = 13.55 * 12 / 4 = 40.65$ kip-ft.

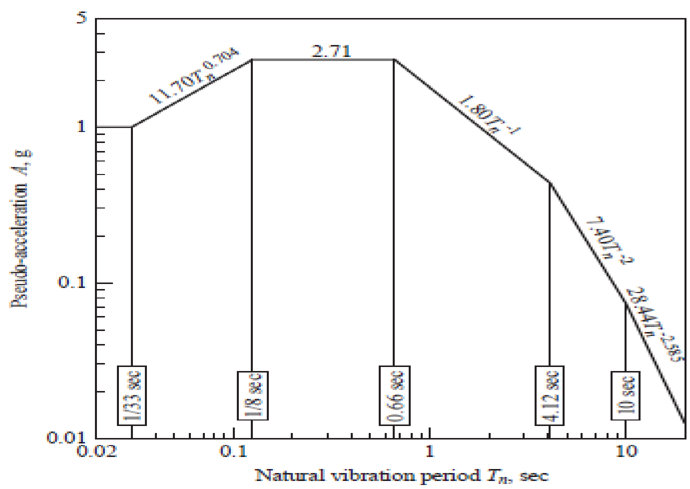
(b) Frame with flexible beam (EI = 0)

k = (2 * 3 * E * I_c) / h^3 = 5.02 kips/in.

$\omega_n = \sqrt{(k / m)}$ = 13.92 rad/sec;

$T_n = 2\pi / \omega_n$ = 0.452 sec;

For T_n = 0.452 sec and ζ = 5%,



A/g = 1.355

A/g = 1.355 => D = A/ $(\omega_n)^2$ = (1.355 * 386) / $(13.92)^2$ = 2.7 in.

Design quantities:

u_o = 2.7 in.

f_{so} = (A / g)w = 13.55 kips

The bending moment at the base of columns is

M = (1355 * 12) / 2 = 81.30 kip-ft

A rigid beam increases the lateral stiffness by a factor of 4 and shortens the natural period by a factor of 2. In the constant-A region of the spectrum, this change in Tn does not affect the lateral force; however, the maximum bending moment is halved. The design deformation is reduced by a factor of four.